

Recursive Binary Splitting

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Overview of Decision Trees

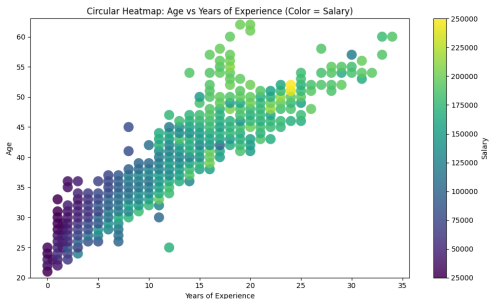
- Decision trees stratify or segment the predictor space into a number of simple regions.
- To make predictions, we use the mean (for regression) or mode (for classification) of observations in each region.
- Tree-based methods are simple, interpretable, and applicable to both regression and classification problems.

Combining Trees

- Single trees are often less accurate compared to more advanced supervised learning methods.
- Combining multiple trees can improve predictive performance.

Example: Salary Data (Kaggle)

- Dataset: Salary Data (Kaggle)
- Response variable: $Y = \text{Salary}$
- Predictors:
 - Age
 - Years of Experience



- **Terminal nodes or leaves:** Final regions $R_1, R_2, \dots, R_J$.
- **Internal nodes:** Points along the tree where the predictor space is split are referred to as internal nodes.
- **Branches:** Segments of the trees that connect the nodes are called branches.

Prediction with a Tree

- Step 1: Divide the predictor space into J distinct, non-overlapping regions $R_1, R_2, \dots, R_J$
- Step 2: For every observation in region R_j , make the same prediction:

$$\hat{y}_{R_j} = \frac{1}{|R_j|} \sum_{i: x_i \in R_j} y_i$$

Dividing the Predictor Space

- The predictor space could theoretically be divided into arbitrary shapes.
- In practice, we divide it into multi-dimensional rectangles (“boxes”).
- This simplifies computation and allows graphical tree representation.

Regression Tree Objective

- The quality of a regression tree is measured by the Residual Sum of Squares (RSS):

$$\text{RSS} = \sum_{j=1}^J \sum_{i: x_i \in R_j} (y_i - \hat{y}_{R_j})^2$$

- Goal: find the partition $R_1, \dots, R_J$ that minimizes the RSS.

Recursive Binary Splitting

- It is computationally infeasible to consider every possible partition of the feature space into J boxes.
- Instead, we use **Recursive Binary Splitting**.
- Formally, we select the predictor X_j and cutpoint s that minimize:

$$\sum_{i: x_i \in R_1(j, s)} (y_i - \hat{y}_{R_1(j, s)})^2 + \sum_{i: x_i \in R_2(j, s)} (y_i - \hat{y}_{R_2(j, s)})^2$$

where:

$$R_1(j, s) = \{X \mid X_j < s\}, \quad R_2(j, s) = \{X \mid X_j \geq s\}$$

Top-Down Greedy Approach

- **Top-down:** it begins at the top of the tree where all observations belong to a single region. Then it successively splits the predictor space.
- **Greedy:** At each step, the best split is made at that particular step. Does not look ahead and picks a split that will lead to a better tree in some future step.

Stopping Criterion

- Continue splitting until a stopping criterion is met.
- Example: stop when no region has more than five observations.
- Without stopping, tree may perfectly fit training data (overfitting).

Overfitting and Complexity

- Trees can easily overfit training data even with some stopping criteria.
- A smaller, simpler tree may have:
 - Lower variance
 - Better interpretability
 - Cost: Slightly higher bias
- One way is to stop early when improvement in RSS falls below a threshold.

Code Demonstration

We now demonstrate the Recursive Binary Splitting algorithm on the Salary dataset.

Thank You!